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SMART HOSTEL MANAGEMENT SYSTEM

1. Project Title and Business Idea

Project Title: Smart Hostel Management System

Business Idea: The Smart Hostel Management System is a software application designed to simplify hostel operations by managing customers, rooms, bookings, payments, and maintenance requests through a centralized database.

2. Problem Statement and Purpose

Traditional hostel management often rely on manual records, which can cause booking errors, misplaced information, difficulties tracking payments, and delays in handling maintenance problems.

The purpose of the proposed system is to improve efficiency, reduce paperwork, minimize errors, and make hostel information easier to manage and retrieve.

3. Project Scope

Functions Covered

- Customer registration and management.
- Room management and availability checking.
- Room booking and reservation management.
- Recording and tracking payments.
- Submitting and tracking maintenance requests.
- Administrator management of rooms, bookings, and maintenance.

Functions Excluded

- Employee payroll management.
- Food and catering services.
- Transportation management.
- Automated door locks and biometric access control.
- Direct integration with banks or mobile payment providers.

4. Main Classes and Their Responsibilities

Class	Responsibility
Customer	Registers customers, books rooms, and makes payments.
Admin	Manages rooms, oversees bookings, and handles maintenance requests.
Room	Stores room details, prices, and availability status.
Booking	Records room reservations and booking dates.
Payment	Stores payment amounts, dates, methods, and statuses.
MaintenanceRequest	Records maintenance problems and tracks their status.

5. System Requirements

Hardware Requirements

- A computer or laptop.
- At least 4 GB of RAM.
- Sufficient storage space for the database and project files.

Software Requirements

- Operating system: Windows, Linux, or macOS.
- PostgreSQL for database management.
- pgAdmin for executing SQL queries and managing the database.
- Draw.io (diagrams.net) for creating and editing the UML class diagram.

Functional Requirements

The system must allow users to manage customer information, rooms, bookings, payments, and maintenance requests.

Non-Functional Requirements

The system should be reliable, secure, easy to use, and capable of storing and retrieving information accurately.

6. Conclusion

The Smart Hostel Management System aims to make hostel operations more efficient by centralizing important records and simplifying daily management activities. PostgreSQL will be used to store and manage the data, while the UML class diagram will guide the system's design.